

# A manifest of docs/reports

One hundred and nine independent mathematical research packages,  
classified by subject

Vladimir Reshetnikov

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## 1 Scope and conventions

This document catalogues the one hundred and nine research packages stored under `docs/reports`. They arrived as one hundred and twenty-five ZIP archives, in seven deliveries, which have been unpacked and grouped into the subject hierarchy described in Section 2. The archives themselves are no longer kept in the repository.

One hundred and twenty-five archives yield one hundred and three reports for two different reasons. Twenty archives duplicated another: sixteen pairs, three three-way clusters and one four-way cluster. Two more were folded into a neighbour not because they repeated it but because they shared a subject — the three reports on term counts in the derivatives of power towers are now one article with a common framework and three parts. That consolidation is editorial, and the distinction matters: those three were put through the duplicate sweep and cleared it, and their three theorems are three different theorems. The seventh delivery of eighteen contained no duplicate at all, which is worth recording: a sweep that finds nothing is evidence about the collection, not a sweep that failed. Each pair has been merged into one report that proves the shared theorem once and keeps everything both originals built on top of it. Where the two reached a result by genuinely different routes, both proofs are kept and marked as alternatives, with a sentence on what each one buys; the merged report on surreal option graphs keeps two different five-vertex counterexamples for the same reason, and the merged shuffle report keeps three independent finite certifications of one theorem. A merged entry below names both of its source archives and says what was doubled.

Not every pair aimed at one problem is a duplicate. The two Gonshor reports both prove his product-birthday bound, but on domains that are incomparable in both directions, so neither theorem contains the other and they are catalogued separately; the same holds for reports answering different numbered problems of one paper, noted where it occurs.

Every package is self-contained and follows the same shape: a typeset article (`*.pdf`) with its complete  $\text{\LaTeX}$  source, a `README.md` summarising the result, usually a `code/` directory with exact-arithmetic verification scripts, a `data/` or `results/` directory holding the recorded output of those scripts, and often `STATUS.md`, `sources.md` or `notes/` files auditing the sources consulted and the precise limits of the claim.

Two cautions apply uniformly and are therefore not repeated in the individual entries. First, these are AI-assisted research drafts: none has been independently refereed, and none has been checked in a proof assistant. Second, the descriptions below record *what each report sets out to settle and what it claims to prove*. Compiling this manifest involved reading each package’s statement of its problem and result; it did not involve checking any proof. Where a report itself flags a limitation — a prior disproof, an imported theorem, a correction to the source, or an exception left open — that is noted, because it changes what the report is.

The naming of the directories is editorial. Several distinct reports attack the same published problem, and some arrived under colliding archive names, so each directory has been given a name describing its own angle on the problem; the merged directories are named for their combined scope, which is why none of them still carries the name that once distinguished it from its partner. The original archive names are recorded with every entry.

## 2 The directory tree

| Category                             | Subcategory                       | Reports    |
|--------------------------------------|-----------------------------------|------------|
| automata-and-formal-languages        | —                                 | 6          |
| ordinals-and-order-types             | friendly-order-types              | 3          |
|                                      | wqo-powersets-and-statutes        | 4          |
|                                      | transfinite-words                 | 2          |
|                                      | ordinal-arithmetic                | 4          |
|                                      | games-on-ordinals                 | 3          |
|                                      | (top level)                       | 4          |
| surreal-numbers                      | —                                 | 6          |
| hankel-determinants                  | somos-and-elliptic                | 3          |
|                                      | catalan-and-ballot                | 4          |
|                                      | growth-and-runs                   | 2          |
|                                      | (top level)                       | 1          |
| congruences-and-valuations           | supercongruences                  | 2          |
|                                      | iterated-series                   | 2          |
|                                      | (top level)                       | 3          |
| tetration-and-digit-stabilization    | —                                 | 6          |
| log-concavity-and-unimodality        | —                                 | 12         |
| graph-theory                         | —                                 | 3          |
| enumerative-combinatorics            | —                                 | 19         |
| generating-functions-and-asymptotics | binary-product-coefficient-counts | 3          |
|                                      | oeis-sequence-asymptotics         | 7          |
|                                      | (top level)                       | 4          |
| <b>Total</b>                         |                                   | <b>103</b> |

More than a quarter of the collection consists of counterexamples and corrections rather than proofs. Ordinal Chomp, open-query games, compositional tree series, the secant numbers, the decimal-length bound for tetration stabilization, power-tower stabilization, iterated-power digit stabilization, cluster variables, chromatic coefficients of cycles, products of  $q$ -integers, domination roots, Hardinian array constants, the mixed-base counterexample, recursive exponents, Gregory sign thresholds and binary substitution discrepancy all refute, or replace with a proved statement, an assertion published or catalogued elsewhere. Several reports whose main outcome is a proof still correct the statement they prove: the Somos-8 certificates, the shifted Catalan Hankel polynomials and the Rueppel determinants record a sign error, a missing one-unit shift and a numbering ambiguity in their sources, and the independence-system thresholds report refines its target after finding that the original conjecture had already been disproved by someone else.

## 3 Ordinals and order types

### 3.1 Friendly order types

Three reports on Isa Vialard’s *friendly order type*  $f(P)$ , the rank of the tree of open-ended bad sequences in which every chosen point has an incomparable partner in the current residual. All three prove the same finite formula; they differ in what they build on top of it, and in how they

prove it — one by a forest-rank monovariant that survives moves deleting a whole upset, one through a lemma bounding the cut maxima. A fourth report proved the formula by the first report’s argument almost verbatim and has been merged into it.

### 1. Friendly Order Types Through Incomparability Components

[ordinals-and-order-types/friendly-order-types/incomparability-components-and-infinite-tails/friendly\\_order\\_types.pdf](#)  
 from [friendly\\_order\\_type\\_research](#) (1).zip

The same component formula, together with an exact friendly rank for connected infinite well-partial-orders: writing  $o(C) = \lambda + n$  with  $\lambda$  the limit part and  $T(C)$  the finite set of points with finite principal upsets. Includes an exhaustive finite verification and a proof audit.

### 2. Friendly Order Type Through Incomparability Components

[ordinals-and-order-types/friendly-order-types/matroid-rank-and-ordinal-sums/article.pdf](#)  
 from [friendly\\_order\\_type\\_research](#) (3).zip

Identifies  $|P| - c(P)$  with the graphic-matroid rank and the incidence-matrix rank of the incomparability graph. Adds a substitution formula, a finite Cartesian-product formula, and additivity over arbitrary well-ordered ordinal sums; evaluates every well-partial-order whose incomparability components are finite, and gives countable counterexamples delimiting stronger infinite extrapolations.

### 3. Friendly Order Types of Finite Posets and Ordinal Products

[ordinals-and-order-types/friendly-order-types/components-forests-and-ordinal-products/friendly\\_order\\_types.pdf](#)  
 from [friendly\\_order\\_types\\_research.zip](#) + [friendly\\_order\\_type\\_research.zip](#)

Proves  $f(P) = |P| - c(\text{Inc}(P))$  for a finite poset  $P$ , where  $c$  counts connected components of the incomparability graph (isolated vertices included), and classifies the witnesses: every maximum friendly subset is  $P$  minus one minimal root per component, counted by  $\prod |\text{Min}(C_i)|$ . With Vialard’s width-transfer theorem this gives ordinal width  $\omega^{f(P)}$  for the Dershowitz–Manna multiset ordering. On the transfinite side it proves  $f(\alpha_1 \times \cdots \times \alpha_r \times B) = \Theta$ , the natural product, except that it is the predecessor of  $\Theta$  when every  $\alpha_i$  is a successor and  $B$  has a greatest element, and classifies arbitrary finite ordinal boxes. Also: spanning-forest certificates, exact monotonicity under added comparisons, a bivariate generating function, and explicit four-point factors with equal  $(o, h, w, f)$  whose products differ. *Merged; the Cartesian-product formula is proved twice, once by corner deletion and once by a chain-grid spanning-subgraph argument.*

## 3.2 Well-quasi-ordered powersets and statures

Four reports on the open questions in Section 6 of Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard, *Measuring well-quasi-ordered finitary powersets* (arXiv:2312.14587v2), and on the corresponding topological invariant.

### 4. Cofinal strata and ordinal absorption

[ordinals-and-order-types/wqo-powersets-and-statures/cofinal-strata-of-finitary-powersets/article.pdf](#)  
 from [cofinal\\_strata\\_ordinal\\_powersets.zip](#)

For  $S_\alpha(Q)$ , a finite poset  $Q$  with each point replaced by a chain  $\alpha = \omega^\rho$ , proves  $o(K(S_\alpha(Q))) = \sum_k \omega^{\rho \otimes k} c_k(Q)$  and  $h(K(S_\alpha(Q))) = \alpha \cdot \text{height}(M(Q))$ , where  $c_k$  counts size- $k$  maximal antichains not dominated by a larger one. The maximal-order-type theorem allows nonuniform pure well-partial-order fibers and is proved by a finite cofinal-stratification argument.

### 5. Intrinsic ordinal approximations for finitary powersets

[ordinals-and-order-types/wqo-powersets-and-statures/intrinsic-ordinal-approximations/article.pdf](#)

from `intrinsic_ordinal_approximations.zip`

Studies the intrinsic invariant  $a_o(A) = \sup\{o(B) + 1 : \text{Idl}(B) \hookrightarrow A\}$ . Proves the universal bound  $a_o(A) \leq h(P_f(A))$  with equality exactly when every finitely generated downset of  $A$  is an algebraic depo, gives the closed formula  $h(P_f(\prod_i \alpha_i)) = \sup(\bigotimes_i (1 + a_i) + 1)$ , and shows the universal identity fails on  $\lambda + F$ .

## 6. Ordinal Order Maps and Finitary Powersets of Ordinal–Finite Grids

`ordinals-and-order-types/wqo-powersets-and-statures/  
ordinal-order-maps-and-grids/ordinal_order_maps.pdf`  
from `ordinal_order_maps_research.zip`

Exact maximal order types and heights for finite Hoare powersets of  $\beta \times Q$  with  $Q$  a finite poset: for  $\beta = \omega^{\gamma_0} + \dots + \omega^{\gamma_{m-1}}$ ,  $o(\text{Mon}(Q, \beta)) = \bigoplus_{s \text{ isotone}} \omega^{\bigoplus_q \gamma_{s(q)}}$ .

## 7. Sharp ordinal bounds for Hoare powerspaces

`ordinals-and-order-types/wqo-powersets-and-statures/  
hoare-powerspace-statures/article.pdf`  
from `hoare_statures_research.zip`

For all Noetherian spaces,  $1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}$ , with witnesses attaining both endpoints at every ordinal and  $T_1$  upper witnesses; the older  $\omega^\alpha$  bound is attainable at infinite input stature  $\alpha$  exactly when  $\omega \cdot \alpha = \alpha$ ; the complete spectrum at  $\omega + n$  is  $\omega \cdot j(U) + (j(Q) - j(U))$ . Extends the known well-quasi-order results of Abriola et al. to arbitrary Noetherian spaces.

## 3.3 Transfinite words

Two reports on Harry Altman, *Bounding finite-image sequences of length  $\omega^k$*  (arXiv:2409.03199v2), Example 3.15: is the bound  $\omega^{\omega^4}$  sharp for words of length  $< \omega^2$  over the two-element antichain? A third report matched the first lemma for lemma and has been merged into it.

## 8. Exact maximal order types of finite-alphabet transfinite words

`ordinals-and-order-types/transfinite-words/  
finite-alphabet-transfinite-words/article.pdf`  
from `finite_alphabet_transfinite_words.zip`

Proves  $o(W_k(P)) = \omega^{\omega^{N_k(P)-1}}$  for words of ordinal length  $< \omega^k$  over a finite poset  $P$  under subsequence embedding, where  $N_k$  counts indecomposable classes of a canonical finite atom poset  $I_k(P)$ , with  $N_1(P) = |P|$  and  $N_{k+1}(P) = |P| + J(I_k(P)) - 1$ . For  $k = 2$  the binary antichain gives  $\omega^{\omega^4}$  — the tightness Altman leaves open — and the binary chain gives  $\omega^{\omega^3}$ .

## 9. Guarded periodic blocks and exact maximal order types below omega squared

`ordinals-and-order-types/transfinite-words/order-types-below-omega-squared/article.pdf`  
from `omega2_order_types_proposed_solution.zip` + `ordinal_words_research.zip`

Proves  $o(s_{\omega^2}(P)) = \omega^{\omega^{n+j(P)-2}}$  for  $n = |P| \geq 2$ , with  $j(P)$  the number of downsets including the empty one; the empty and singleton alphabets are genuine exceptions. The binary antichain gives  $\omega^{\omega^4}$  — the tightness Altman leaves open — and the binary chain  $\omega^{\omega^3}$ . Adds an extremality corollary with both equality characterisations (among  $n$ -element posets only the chain minimises and only the antichain maximises), the  $j(P)$  toolkit, the arbitrary-support-family classification, and an exact-length stratum calculus showing the supremum over lengths is only  $\omega^{\omega^n}$ , strictly below the answer — so the theorem cannot be obtained stratum by stratum. *Merged; the product amplification is done twice, once via de Jongh–Parikh and once by an elementary hand proof that removes the import.*

## 3.4 Ordinal arithmetic

Four reports on ordinal and infinitary arithmetic, two of them independent proposed solutions to Problem 6.2 of Paolo Lipparini, *Monotone infinitary operations on ordinals* (arXiv:2505.00424v2):

define and study the least weakly monotone,  $e$ -special strictly increasing operation on countable sequences of ordinals.

### 10. An Explicit Formula for the Minimal $e$ -Monotone Infinitary Ordinal Sum

[ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-infinitary-sum/article/explicit\\_ordinal\\_sum.pdf](#)

from [lipparini\\_problem\\_6\\_2\\_proposed\\_solution.zip](#)

An explicit formula for the least admissible operation, proved by an arithmetic admissibility upper bound and a rank-theoretic minimality lower bound, using an explicit countable-multiset profile order, its rank, absorption and correction rules, bounded-poset heights and finite-cap ranks. Imports Lipparini's formula and minimality theorem for the weaker operation  $S$ .

### 11. An Explicit Formula for Lipparini's Minimal Infinitary Ordinal Operation

[ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-operation-formula/article.pdf](#)

from [ordinal\\_operation\\_research.zip](#)

A three-case finite ordinal-arithmetic formula with proofs of admissibility and pointwise minimality. Relative to Lipparini's already evaluated  $S$ , the correction is zero, a finite increment, or one *natural* summand  $\omega$  — ordinary addition of  $\omega$  is not a substitute.

### 12. Jacobsthal Products All at Once

[ordinals-and-order-types/ordinal-arithmetic/jacobsthal-products-all-at-once/article.pdf](#)

from [Jacobsthal\\_Products\\_All\\_at\\_Once.zip](#)

Question 2.16 of Altman, *Intermediate arithmetic operations on ordinal numbers* (MLQ 63 (2017) 228–242), asks for an all-at-once order-theoretic model of finite and infinitary Jacobsthal products, rather than one built from parenthesized binary multiplication or a limit. Gives a constructive affirmative answer: a Cantor-normal-form comparison rule on actual tuples or finite-support functions, invariant under order isomorphisms and extending the componentwise order, whose definition never evaluates a Jacobsthal product. Adds a restricted maximality characterisation and a finite-tail reduction. The associativity identity and the binary normal-form formula are classical and are not claimed here; the construction is not claimed to be preserved by arbitrary order embeddings.

### 13. Pairwise Coherence Does Not Guarantee Completion

[ordinals-and-order-types/ordinal-arithmetic/pairwise-coherence-of-infinitary-products/ordinal\\_product\\_coherence.pdf](#)

from [ordinal\\_product\\_coherence.zip](#)

Answers the pairwise-coherence sufficiency question of Lipparini, *Noncommutative infinitary semigroups* (arXiv:2605.28413v1, Remark 7.1(b)), negatively. On the two-element group  $C_2$ , retaining every finite product and declaring every infinite all- $e$  product to be  $e$  gives a partial system satisfying full pairwise regrouping, even for arbitrary partitions, yet with no injective extension defining all finite and  $\omega$ -indexed products; two elements is minimal. Adds: for any monoid  $M$  the canonical partial domain  $P(M)$  is coherent, and for directly finite  $M$  it completes exactly when the unit group is trivial; and for every  $r$  a system on  $3r + 4$  elements coherent through nesting depth  $r$  but conflicting at depth  $r + 1$ , so no fixed finite depth of the coherence test characterises completability.

## 3.5 Games on ordinals

### 14. An ordinal Chomp transition at 2

[ordinals-and-order-types/games-on-ordinals/ordinal-chomp-transition-at-two/ordinal\\_chomp\\_counterexample.pdf](#)

from [ordinal\\_chomp\\_transition\\_counterexample.zip](#) + [ordinal\\_chomp\\_counterexample.zip](#)

For a numerical semigroup  $S$  ordered by  $x \leq_S y$  iff  $y - x \in S$ , and the ordinal extension ordered by ordinary ordinal subtraction (so  $S^2$  is the *lexicographic*, not coordinatewise, square), Chomp on  $S$

is a second-player win while Chomp on  $S^\sigma$  is a first-player win for every  $\sigma \geq 2$ , and  $\omega \cdot 4 + 4$  is the *least* winning opening. Hence  $\text{ch}(\langle 4, 5, 6 \rangle) = \text{ch}(\langle 4, 6, 9 \rangle) = 2$ , a negative answer to Question 4.6 of Rivero Herrera (arXiv:2504.07317v3) and a refutation of Conjecture 4.1 of v1, already for three small natural-number generators. The proof reduces the infinite game to a finite spectral statement via a finite-window recurrence and an exact finite-state certificate, then transfers it by an explicit response procedure. *Merged; the shared  $\langle 4, 6, 9 \rangle$  certificates were verified to hold the same 253 rows entry for entry, and the two reports' poisoned and unpoisoned conventions are reconciled by an explicit dictionary.*

### 15. Every ordinal as a point-separating game value

[ordinals-and-order-types/games-on-ordinals/  
point-separating-game-values/ordinal\\_separation.pdf](#)  
from ordinal\_separation\_research.zip

For every ordinal  $\lambda$ , orders  $\{0, 1\}^\lambda$  lexicographically and gives it the topology of all downward-closed sets, and the refinement generated by downward-closed sets minus finite sets. The point-separating game value is exactly  $\lambda$  for both: the first space is compact  $T_0$ , the second compact  $T_1$ , and for infinite  $\lambda$  neither is Hausdorff. Taking  $\lambda = \omega^2 + 1$  supplies a compact  $T_1$  witness of cardinality continuum for Problem 1.2 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3), with no CH, forcing or large-cardinal hypothesis. Hausdorff witnesses are not claimed. This is a different problem of that paper from the one the open-query games report answers.

### 16. Open-query games on ordinal spaces

[ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.pdf](#)  
from ordinal\_membership\_games.zip

A negative answer to Problem 1.3 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3): two copies of  $[0, \omega]$  each have set-membership number 1, but their topological sum has value 2. The general theorem computes the invariant for every nonempty Hausdorff space of finite Cantor–Bendixson height  $h$  as  $\lceil \log_2(h + 1) \rceil$  when the last nonempty derivative is a singleton and  $\lceil \log_2(h + 2) \rceil$  otherwise.

## 3.6 Order-theoretic invariants without a subcategory

Two reports whose objects sit outside every subcategory above: the minor order on finite graphs, and a site of filters on  $\omega$ .

### 17. Ordinal complexity of graph minors

[ordinals-and-order-types/graph-minor-ordinals/graph\\_minor\\_ordinals.pdf](#)  
from graph\_minor\_ordinals.zip

For finite simple graphs of maximum degree at most two under minor containment, computes maximal order type, ordinal antichain-tree width and descending-tree height:  $o(D) = w(D) = \omega^{\omega \cdot 2}$  and  $h(D) = \omega$ , with exact values  $\omega^{b+k}$  and  $\omega^{b+k-1}$  for the subfamilies bounded by path size  $b$  and cyclic-component count  $k$ . The general theorem is that a wqo  $C$  of connected graphs with  $o(C) = \alpha < \varepsilon_0$  has finite disjoint-union closure of type  $\omega^\alpha$ , proved by a component-splitting argument, since minor containment is not multiset embedding. A corollary is that continuity fails for increasing unions of minor-closed classes. The report is explicit that it does not compute the maximal order type of the minor order on all finite graphs and settles no named conjecture.

### 18. A two-valued obstruction to gluing Fréchet powers

[ordinals-and-order-types/frechet-two-valued-obstruction/article.pdf](#)  
from frechet\_two\_valued\_obstruction.zip

For the presheaf  $F \mapsto A^\omega / \sim_F$  on proper filters on  $\omega$  containing the cofinite filter, with the dense Grothendieck topology, proves in ZF that it is a sheaf if and only if  $|A| \leq 1$ . The counterexample is an explicit matching family of constant sections using two values, on a covering sieve with countably many generators each of countable base; the distinction between the number of sieve generators and the size of the full sieve is essential to it. The question is Massas, *A Semi-Constructive*

*Approach to the Hyperreal Line* (2023), p. 517 after Fact 5.10, repeated in the 2024 dissertation *Duality and Infinity*, p. 249.

### 19. A sharp cover bound for congruences of finite lattices

[ordinals-and-order-types/lattice-congruence-cover-bound/sharp\\_cover\\_bound.pdf](#)  
 from [lattice\\_cover\\_bound\\_research.zip](#)

If a finite  $n$ -element lattice has an element with at least  $k \geq 3$  upper covers then  $|\text{Con}(L)| \leq 2^{n-k-1}$ , with equality exactly for the glued sums  $C_s \oplus M_k \oplus C_t$ . This replaces the multicolour Ramsey threshold  $R_{k-1}(k)$  of the source's Lemma 4.4 (arXiv:2603.11454) by  $k$  itself. Adds a sharp nonextremal gap, a fan-label ideal refinement, and a bound on the size of Czédli's skeleton in terms of congruence density. All congruence machinery, including Freese's maximum, is proved from scratch. The target is an informal suggestion in the source, not a numbered conjecture.

### 20. A non-Baire translation-invariant ideal

[ordinals-and-order-types/non-baire-translation-invariant-ideal/article.pdf](#)  
 from [non\\_baire\\_ideal\\_research.zip](#)

An affirmative construction for Question 1 of Filipów and Tryba, *Densities for sets of natural numbers vanishing on a given family* (J. Number Theory 211 (2020) 371–382; arXiv:2309.00982, p. 7): a translation-invariant ideal that is not Baire and carries a perfect translation-almost-disjoint family.

## 4 Automata and formal languages

Five reports on state complexity, synchronization and language hierarchies. The first two are each a merge: the shuffle bound arrived four times and the DFAO reversal bound twice.

### 21. The six-state shuffle bound

[automata-and-formal-languages/shuffle-six-state-bound/shuffle\\_six\\_state.pdf](#)  
 from [shuffle\\_six\\_state\\_research \(1\).zip](#) + [shuffle\\_six\\_state\\_research.zip](#) +  
[Shuffle\\_Six\\_State\\_Theorem.zip](#) + [six\\_state\\_shuffle\\_research.zip](#)

For complete DFAs with  $m, n \geq 2$  and  $\min(m, n) \leq 6$ , the worst-case deterministic state complexity of the shuffle is  $F(m, n) = 2^{mn-1} + 2^{(m-1)(n-1)}(2^{m-1} - 1)(2^{n-1} - 1)$ , over a finite alphabet allowed to depend on  $m$  and  $n$ ; the unrestricted conjecture stays open. The route is a reachability theorem in the universal shuffle automaton — a subset is reachable exactly when it meets the initial row and the initial column — proved by induction on  $(m + n, |S|)$  down to a finite certificate obligation. *Merged from four reports. Two shipped the same 10,642 witnesses; a third supplies an independent database of 10,619 witnesses sharing no witness with them, so the finite step is certified twice over; a fourth adds a fifth analytic reduction that cuts the obligation to 453 certificates and replaces the solver search with a deterministic enumeration. All three routes are kept.*

### 22. A coloring obstruction to maximal binary DFAO reversal

[automata-and-formal-languages/dfa-reversal-coloring-obstruction/article.pdf](#)  
 from [dfa-reversal\\_research.zip](#) + [binary\\_dfa-reversal\\_research.zip](#)

For transformations  $a, b$  on an  $n$ -element set and an output map to  $k$  labels, proves  $|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n$  for  $3 \leq k \leq n$ , with  $g(k)$  the largest order of a permutation of  $k$  symbols — the binary-alphabet nonattainment question of Davies, arXiv:1705.07150v2, Section 5. The mechanism is a collision-orbit graph, a rightmost singular factorization, and a cyclic relabeling bound. Adds an abelian-permutation extension, a structural universal bound matching the published lower construction on all 372 pairs with  $7 \leq n \leq 30$ , and polynomial-time missing-coloring certificates. *Merged; the two originals ran the same three-case argument, and both a qualitative short proof and the counting proof are kept.*

### 23. Synchronizing Dzyga's automata by a cyclic retract

[automata-and-formal-languages/dzyga-synchronizing-retract/article.pdf](#)

from Dzyga\_Automata\_Cyclic\_Retract\_Research.zip

For the Figure-3-complete automaton  $A_k$  of Conjecture 8 in Szykula's 2026 survey, exhibits an explicit word  $R_k = p(ec^{r-1})^{r-2}e$  with  $r = k + 4$  that sends every state to  $q_2$  for every  $k \geq 1$ , of expanded length  $8k^2 + 54k + 92 = \frac{8}{9}n^2 + \frac{22}{3}n + 16$  for  $n = 3k + 6$ . The proof is a cyclic retraction with two projected reflections whose discrepancy supplies one adjacent merger, and does not depend on the finite computations; shortening the reflected tail by one state gives a nonsynchronizing automaton with a dihedral permutation quotient. This settles the synchronization clause of the stated model, not the numerical threshold conjecture, and the universal lower bound  $\tau(q_0) \geq 4k + 8$  is not proved.

#### 24. Gaps in bounded-shuffle hierarchies

automata-and-formal-languages/insertion-degree-spectra/article.pdf

from shuffle\_spectra\_research.zip

A counterexample to the insertion-degree No-Gap Conjecture of Hughes, arXiv:2608.27755v1, Section 11, followed by positive results: every finite set of positive integers containing 1 is the insertion-degree spectrum of two nonempty finite languages over a suitable alphabet, and any prescribed finite list of degrees above 1 is realizable with every other output of degree 1. A ternary pair has spectrum  $\{1, 3\}$  with sole difficult output **abcba**, whose length 5 is the minimum possible for a gap witness; a binary pair has spectrum  $\{1, 2, 4\}$  while all 18 singleton source-pair spectra are initial intervals, so source-pair ambiguity can erase a degree. The separate iteration-depth no-gap conjecture is proved for arbitrary languages.

#### 25. A cut-sensitive attack on the first open MIX grammar case

automata-and-formal-languages/mix-grammar-cut-sensitive/mix\_grammar\_report.pdf

from mix\_grammar\_research.zip

The selected question — whether  $L_3 = \text{MIX}$  for the canonical tuple grammars of Yuyama, *On the Kanazawa–Salvati Conjecture* (arXiv:2609.13871v1) — is *not* settled here, and the report says so at the front. What it does contain, with complete written proofs, is a cut-sensitive membership recurrence, a broad three-component positive family, an infinite family in  $L_3 \setminus L_2$ , and an exhaustive computation through length 21.

#### 26. Natural densities of the Tribonacci word's additive complexities

automata-and-formal-languages/tribonacci-additive-complexity/article.pdf

from tribonacci\_additive\_density\_research.zip

Answers Remark 19 of Popoli, Shallit and Stipulanti (arXiv:2410.02409). With  $\beta$  the Tribonacci constant, the additive complexity of the Tribonacci word takes values in  $\{3, 4, 5\}$  with natural densities  $d_3 = (21\beta^2 - 38\beta + 5)/22$ ,  $d_4 = (-55\beta^2 + 72\beta + 67)/22$  and  $d_5 = (17\beta^2 - 17\beta - 25)/11$ , all positive and summing to 1, with a power-saving error  $O(N^\theta)$ ,  $\theta = \log(7/5)/\log \beta$ , at *every* integer cutoff rather than only along Tribonacci cutoffs. The 76-state output automaton is reconstructed independently rather than copied, and certified by evaluating its degree-76 characteristic polynomial at the adjacency matrix.

## 5 Surreal numbers

Six reports on the surreal field itself: birthdays and their behaviour under multiplication, the option graphs of surreal forms, genetic functions and gaps, and two questions of Lipparini about evaluating formal sums in the surreals. The first four arrived as two pairs; each pair was merged.

#### 27. A square-root obstruction to evaluating increasing Hahn series at $\omega$

surreal-numbers/hahn-evaluation-at-omega/article.pdf

from hahn\_evaluation\_research.zip + reversed\_hahn\_series.zip

Answers Problem 7.7 (p. 38) of Lipparini (arXiv:2505.00424v2) negatively: there is no ring homomorphism from the field of set-supported real-coefficient Hahn series with strictly increasing surreal exponents into the surreals sending  $x^a$  to  $\omega^a$ . The certificate is  $B \in \mathbb{Q}[[x]]$  with  $B^2 = 1 - x$ ,

built by the recursion  $b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k}$ : any such map would force a square to equal  $1 - \omega < 0$ . The homomorphism is applied only to the finite identity, so no continuity, valuation, order preservation or preservation of infinite sums is assumed. Adds the exact classification of admissible images of the variable as the infinitesimals, the monomial classification over an arbitrary ordered abelian group, and order-uniqueness on the source field. *Merged; the core certificate was identical in both originals and is proved once, but the two toolkits are kept — in particular a nonnegative-coefficient semiring obstruction with strictly weaker hypotheses that the square-root certificate cannot reach.*

## 28. Gonshor’s Product-Birthday Conjecture for ordinal-support series

[surreal-numbers/gonshor-product-birthdays/surreal\\_product\\_birthdays.pdf](#)

from [gonshor\\_product\\_birthdays\\_research.zip](#)

Proves Gonshor’s bound  $b(xy) \leq b(x) \otimes b(y)$  on the ring of Conway normal forms  $\sum_{\alpha \in S} c_\alpha \omega^{-\alpha}$  with  $S$  an arbitrary set of ordinals, so transfinite and uncountable supports are allowed. Supplies an exact birthday formula by transfinite induction over the support with maintained prefix bounds and an explicit limit stage, and a complete classification of the equality cases. The route is a leading-Cantor-degree gap, which also gives that the degree of the birthday never rises under multiplication. The conjecture for arbitrary surreal numbers is not solved. The domain is incomparable with that of the Laurent-series report that follows: it excludes  $\omega$  and every positive power.

## 29. A Laurent-series case of Gonshor’s birthday conjecture

[surreal-numbers/gonshor-laurent-birthdays/article.pdf](#)

from [Gonshor\\_Laurent\\_Birthday\\_Research.zip](#)

Proves the same bound  $b(xy) \leq b(x) \otimes b(y)$  on the canonically embedded field  $\mathbb{R}((\omega^{-1}))$ , which unlike the ring of the preceding entry contains  $\omega$  and its positive powers but restricts supports to order type at most  $\omega$  on the integer lattice. The route is a bilinear budget: an exact splitting  $b(x) = b(P) \oplus b(u)$  into polynomial and infinitesimal parts, three separate product estimates for the resulting cross terms, and a finite-jet stabilisation carrying the finite bound to infinite series without any continuity claim. Also treats finite normal forms with arbitrary ordinal exponents, derives exact birthdays for finite and infinite Laurent series, and proves a reciprocal corollary. Neither this theorem nor the preceding one contains the other; on their common corner  $\mathbb{R}[[\omega^{-1}]]$  the two agree and are proved twice.

## 30. Canonical forms need not be subgraphs

[surreal-numbers/canonical-forms-need-not-be-subgraphs/surreal\\_graphs.pdf](#)

from [surreal\\_graphs\\_research.zip](#) + [surreal\\_sparse\\_graphs.zip](#)

Answers negatively the question in the Figure 3 caption of Roughan, *Surreal Birthdays and Their Arithmetic* (arXiv:1810.10373v2, p. 7): the canonical form of a surreal value need not be a subgraph of every other form of that value. Five vertices and rank three are both least possible, and there are exactly sixteen five-vertex counterexamples, eight triangle-free, four of value  $1/2$ . Adds cycle representations of  $1/2$  at every admissible length, the exact extremal function, a planar construction of maximum degree three and arbitrarily large girth, classifications of which canonical graphs are unavoidable, and transfinite witnesses. *Merged; the two originals exhibited genuinely different five-vertex witnesses — one passing through a vertex of value  $-1$ , one with no negative value at all — and both are printed in full, as are two odd-cycle families, two high-girth constructions and two proofs of the Fibonacci bound.*

## 31. A countable genetic cut across the finite–infinite gap

[surreal-numbers/genetic-gaps-and-primitives/article.pdf](#)

from [genetic\\_gap\\_counterexamples.zip](#)

With  $q(t) = t/(1+t^2)$  and  $B(x) = 1 - \{q(n-x) : n \in \mathbb{N} \mid \}$  as a Conway cut, proves that  $B$  is the indicator of the positive infinite surreals, is genetic under the set-sized option constructors of the cited sources, and has pointwise derivative zero everywhere. Consequently  $x$  and  $x - B(x)$  are genetic primitives of 1 agreeing at every real input that do not differ by a constant, the latter also

an increasing bijection. The sublevel class  $\{x : 0 < x < \omega, B(x) \leq \frac{1}{2}\}$  is nonempty and bounded yet has no surreal supremum, so neither end gap supplies one.

### 32. A Number-Valued Broadcast Sum of Surreal Sequences

[surreal-numbers/broadcast-sum-of-surreal-sequences/article.pdf](#)

from [surreal\\_broadcast\\_sum.zip](#)

Analyses the sign-truncation game of Lipparini (arXiv:2505.00424v2, Remark 7.6(3), in connection with Problem 7.5). Proves the game always has a surreal-number value, that it agrees with finite surreal sums and with Lipparini’s ordinal operation on their respective domains, and that it fails weak coordinatewise monotonicity, with an explicit family  $s[n] = 3 \cdot 2^{-n-d-1}$  on a subset  $E$  and  $2^{-n-d}$  off it realising the failure. The underlying rule is Lipparini’s; the name and the notation are local to the report.

## 6 Hankel determinants and continued fractions

### 6.1 Somos sequences and elliptic curves

#### 33. A proof of Barry’s Hankel–Somos-6 conjecture

[hankel-determinants/somos-and-elliptic/barry-somos-6-hankel-recurrence/article.pdf](#)

from [barry\\_somos6\\_research.zip](#)

Conjecture 7 of Paul Barry, arXiv:2211.12637: for the Hankel determinants  $H_N$  of a four-parameter Jacobi-type generating function,  $H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2$  with  $\alpha = t^2(r+2)^2$  and  $\gamma = t^3((s+t)(r+s+t+2)^2 + t(r+2)^3)$ . Also proves a homogeneous four-parameter form and a seven-parameter polynomial Hankel theorem making exceptional specialisations rigorous; the verifier never divides by a determinant.

#### 34. Finite Hankel Certificates for Three Somos-8 Conjectures

[hankel-determinants/somos-and-elliptic/somos-8-hankel-certificates/article.pdf](#)

from [somos\\_hankel\\_proofs.zip](#)

Conjectures 11, 12 and 13 of the same Barry paper, by importing Hone’s genus-two Hankel theorem (arXiv:1907.05204v3, Thm. 5.4) and supplying a binomial translation, a polynomial-minor specialisation valid on the degenerate locus, and a rank-bound lemma that turns four symbolic rows into an all-index identity. Identifies a sign error in the source’s Catalan expression for Conjecture 13.

#### 35. Elliptic addition and Jacobi continued fractions

[hankel-determinants/somos-and-elliptic/](#)

[elliptic-addition-and-jacobi-fractions/article.pdf](#)

from [elliptic\\_continued\\_fraction\\_research.zip](#)

A consistently indexed, non-torsion version of Barry’s 2023 Conjecture 4, with the coordinate formulas of Conjecture 3: every continued-fraction tail is given explicitly and removing one layer is shown to be elliptic translation. Hypotheses: characteristic zero,  $Y^2 + aXY + bY = X^3 + cX^2 + dX$  nonsingular,  $b \neq 0$ , and  $P = (0, 0)$  of infinite order; order-three and order-four examples show the regular fraction genuinely needs the last one.

### 6.2 Catalan and ballot moments

#### 36. Shifted Catalan Hankel Polynomials

[hankel-determinants/catalan-and-ballot/shifted-catalan-hankel-polynomials/article.pdf](#)

from [catalan\\_hankel\\_conjecture\\_solution.zip](#)

For  $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})$ , proves the denominator-exponent assertion on page 4 of Barry, arXiv:2011.10827v1, and shows the separately printed Conjecture 2 needs an explicit one-unit shift correction. The minimal characteristic polynomial is  $(X^2 - (a+2b)X + b^2)^{m(m-1)/2+1}$ ,

so the minimal order is  $m(m-1)+2$ ; all exceptional reduced denominators, the exact coefficient formula, numerator degree and reciprocal symmetry are supplied.

### 37. Product formulas for ballot-polynomial Hankel determinants

[hankel-determinants/catalan-and-ballot/  
ballot-polynomial-hankel-determinants/article.pdf](#)  
from [Cigler\\_Hankel\\_Conjectures\\_13\\_15.zip](#)

Conjectures 13, 14 and 15 in Section 5 of Johann Cigler, arXiv:2111.14492v3. An explicit coefficient product gives the sign and exact support of the normalised shift- $k$  Hankel determinant of the ballot moments and positivity of every residual coefficient for  $n \geq 1$ ; the same products give the conjectured generating-function denominators, exact numerator degrees and reciprocal numerator identities. Adds strict log-concavity of the residual coefficients and their limiting binomial profile.

### 38. Rectangular Schur polynomials and Cigler's Conjecture 8

[hankel-determinants/catalan-and-ballot/cigler-conjecture-8-schur/article.pdf](#)  
from [cigler\\_conjecture8\\_research.zip](#)

A complete proof of Conjecture 8 in Cigler, arXiv:2111.14492v3, Section 4 (printed pp. 16–17), for the Hankel determinants of  $b_m(t) = \sum_r \binom{\lfloor m/2 \rfloor}{r} \binom{\lceil m/2 \rceil}{r} t^r$ , including the numerator-polynomial degree and reciprocity assertions. A different conjecture of that paper from the ballot-polynomial entry (Conjectures 13–15) and from the parity entry (Conjecture 16).

### 39. A parity factorization for Cigler's Conjecture 16

[hankel-determinants/catalan-and-ballot/cigler-conjecture-16-parity/article.pdf](#)  
from [cigler\\_conjecture16\\_research.zip](#)

A complete proof of Conjecture 16 in Cigler, arXiv:2111.14492v3, Section 6 (pp. 21–22), on the Hankel polynomials, together with a sharp stabilization statement: the report also determines the first coefficient at which the conjectured stabilization fails. Its own README explains why this is a different problem from the other Cigler entries in this collection.

## 6.3 Arithmetic Hankel determinants

The numerator formula recorded as a conjecture in OEIS A122251, with A122249 as its binary special case. Two reports attacked it and proved it the same way, sharing even their worked counterexamples; they are merged here into one. The merged report sits at the category's top level.

### 40. A Proof of an OEIS Hankel-Numerator Conjecture

[hankel-determinants/a122251-numerators-and-denominators/report.pdf](#)  
from [hankel\\_numerator\\_research.zip](#) + [hankel\\_oeis\\_conjecture\\_proof.zip](#)

For  $H_n(m, a) = \det(1/(a+m(i+j)))_{0 \leq i, j \leq n}$ , the reduced numerator is  $\prod_{p|m} p^{n(n+1)v_p(m)+2 \sum_{k \leq n} v_p(k!)}$ : independent of the coprime offset  $a$ , a perfect square, and supported on the primes dividing  $m$ . Taking  $m = p$  and  $a = 1$  settles the OEIS A122251 conjecture, with A122249 as the binary case. Also gives a closed form for the reduced denominator and all its local valuations, a sharp gcd theorem over the allowed offsets realised at just two explicitly constructed shifts, the extension to unequal row and column steps  $\det(1/(a+m_1 i + m_2 j))$ , recurrences, digit-sum asymptotics, and a Lambert-series generating function with a natural boundary. *Merged; the determinant is evaluated three independent ways, and a second proof of the numerator formula runs through an explicit inverse matrix and MacMahon's boxed plane partitions.*

## 6.4 Growth laws and run statistics

### 41. A proof of the Apéry Hankel growth conjecture

[hankel-determinants/growth-and-runs/  
apery-hankel-determinant-growth/apery\\_hankel\\_growth.pdf](#)

from `apery_hankel_research.zip`

For  $A_k = \sum_j \binom{k}{j}^2 \binom{k+j}{j}^2$  and  $D_n = \det(A_{i+j})$ , proves  $\log D_n = n(n+1) \log \frac{17+12\sqrt{2}}{4} + O(n)$  — the limit conjectured in the September 2026 revision of OEIS A228143 — using Edgar’s moment-density theorem. Also an explicit law for  $\det(A_{m(i+j)+r_n})$  with  $r_n/n \rightarrow \rho$ , and non-D-finiteness of the formal generating series of  $D_n$ .

## 42. Binary Runs in Rueppel Hankel Determinants

`hankel-determinants/growth-and-runs/rueppel-binary-run-determinants/rueppel_hankel.pdf`  
from `rueppel_hankel_research.zip`

For the Rueppel series  $r(x) = \sum_{j \geq 0} x^{2^j-1}$ , the polynomial identity  $H_n(tx + 1/r(x^2)) = (-1)^{1+\varepsilon+\lfloor R(n)/2 \rfloor} (\varepsilon + (R(n) - \varepsilon)t^2)$ , where  $R(n)$  counts runs of equal bits in the binary expansion of  $n$  and  $\varepsilon$  is the low bit of the Gray code  $n \oplus \lfloor n/2 \rfloor$ . Settles statements 8, 10, 13 and 23 of Barry’s 2021 paper (not the identically numbered ones elsewhere), with a correction.

# 7 Congruences and $p$ -adic valuations

## 7.1 Supercongruences

## 43. Congruences for OEIS A348410 at every prime

`congruences-and-valuations/supercongruences/a348410-all-prime-congruences/article.pdf`  
from `a348410_supercongruence.zip` + `A348410_Supercongruence_Research.zip`

For  $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$  proves the odd-prime cubic supercongruence  $v_p(A(Np) - A(N)) \geq 3v_p(N) - \varepsilon_p$  with  $\varepsilon_3 = 1$ ,  $\varepsilon_p = 0$  for  $p \geq 5$ , and no coprimality condition;  $(\alpha, \beta) = (2, 1)$  is Bala’s A348410 conjecture. Separately proves an exact binary leading-defect formula,  $A(m2^r) - A(m2^{r-1}) \equiv 2^{3r-3} \alpha \beta \binom{(\alpha+\beta+1)m-1}{m-1} \pmod{2^{3r-2}}$ , hence the all-prime bound  $3r - v_p(12)$  and sharp denominators 12 and 24. An appendix audits arXiv:2104.10754v1, whose framing conclusion and weighted harmonic assertion fail on its own displayed examples. *Merged; the odd-prime and binary halves run on different mechanisms — a doubling automorphism, and reflection with formal integration by parts — and both are kept.*

## 44. Signed binomial supercongruences and OEIS A352373

`congruences-and-valuations/supercongruences/a352373-signed-binomial-supercongruences/signed_binomial_supercongruence.pdf`  
from `OEIS_A352373_supercongruence.zip`

For  $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$  proves  $a(mp^h) \equiv a(mp^{h-1}) \pmod{p^{3h}}$  for  $p \geq 5$  and  $a(m3^h) \equiv a(m3^{h-1}) \pmod{3^{3h-1}}$ , with  $m$  not required coprime to  $p$ . Covers A352373, A348410, A351856 and A351857 as specialisations; the only non-elementary import is the classical Jacobsthal binomial congruence.

## 45. Apéry numbers under exponentiation and series reversion

`congruences-and-valuations/supercongruences/a267220-apery-transform-supercongruences/article.pdf`  
from `Apéry_OEIS_Supercongruences.zip`, `apery_oeis_conjectures(1).zip`

Proves both supercongruence conjectures contributed by Peter Bala to OEIS A267220 on 17 October 2024, for every integer parameter including negative and zero. With  $A(x) = \exp \sum_{n \geq 1} a_n x^n / n$  for the Apéry numbers  $a_n$ ,  $F(x) = x / \text{Rev}(xA(x))$ ,  $u_m(n) = [x^n]A(x)^{mn}$  and  $v_m(n) = [x^n]F(x)^{mn}$ : for odd  $p$  and  $R = \nu_p(N)$  both families satisfy  $\nu_p(u_m(N) - u_m(N/p)) \geq 2R + \nu_p(m)$ , the unsigned statement for  $v_m$  holds at *every* prime including 2, and a *signed* refinement  $(-1)^{mN} u_m(N) \equiv (-1)^{mN/p} u_m(N/p)$  recovers full precision at  $p = 2$ , where the unsigned bound genuinely loses one power. Merged from two reports that share a seed lemma and a Möbius/dilogarithm layer but diverge at the crux; both routes are kept, and the merged text is careful that the signed and unsigned 2-adic statements are different statements, not a sharper and a weaker version of one.

## 46. Four Motzkin congruences

[congruences-and-valuations/supercongruences/a001006-motzkin-congruences/article.pdf](#)  
from [motzkin\\_congruences.zip](#)

Proves all four conjectures in Peter Bala's comment of 10 February 2022 on OEIS A001006, with sharp ranges. The engine is a prime-power block-transfer theorem: for odd  $p$  and  $q = p^{k-s+1}$ ,  $h = mp^{s-1}$ , the Motzkin numbers satisfy  $M_{mp^k+d} \equiv T_h M_d$  for  $0 \leq d \leq q-3$ , with explicit corrections at  $d = q-2$  and  $d = q-1$ ; the boundary case gives  $M_{np^r-2} \equiv \frac{\varepsilon_p^r-1}{2} T_{n-1} - U_{n-1} \pmod{p}$ . Adds the exact first failures beyond the two initial-block ranges, arbitrary-multiplier and all-prime-power boundary extensions, a prime-modulus digit algorithm, and examples showing why unweighted repetition modulo a prime cube is generally false. Discusses Burns, Kohen and Pan-Sun rather than claiming novelty.

## 7.2 Congruences for iterated and compositional series

### 47. Arithmetic of a Self-Iterating Generating Function

[congruences-and-valuations/iterated-series/a396807-self-iterating-series/article.pdf](#)  
from [A396807\\_congruences\\_and\\_prime\\_power\\_lifts.zip](#)

Both conjectures in OEIS A396807 for  $A(x) = x + A^{[5]}(x)A^{[6]}(x)$  (functional iteration):  $A(x) \equiv x/(1-x)$  and  $A^{[k]}(x) \equiv x/(1-kx)$  modulo 10 for every integer  $k$ , negative iterates included. Adds a sharp modulus theorem for  $F = x + cF^{[r]}F^{[s]}$ , rational lifts at every selected prime power, and minimal period 46860 for A396807 modulo 100 starting at index 1.

### 48. Prime-power congruences for compositional tree series

[congruences-and-valuations/iterated-series/compositional-tree-series-congruences/article.pdf](#)  
from [compositional\\_tree\\_congruences.zip](#)

For  $A_l(x) = x \exp(A_l^{ol})$ , shows the OEIS A396805 claim  $a_5(n) \equiv n \pmod{5}$  is *false*: first failure  $n = 23$ , with exception set  $\{20j+3, 20j+4 : j \geq 1\}$  and residue 2 throughout. Proves eleven other conjectural assertions in A396803–A396806 via a general reduction of coefficients mod  $p^\alpha$  to typed rooted trees on fewer than  $p^\alpha$  labels, yielding eventual periods, rational residue generating functions, and an exact classification of  $a_l(n) \equiv n \pmod{p}$ .

## 7.3 Valuations, divisibility and periodicity

### 49. Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers

[congruences-and-valuations/a000139-binary-carries-parity/research.pdf](#)  
from [a000139\\_carries\\_research.zip](#)

Proves Peter Bala's parity conjecture for OEIS A000139,  $a(n) = 2(3n)!/((n+1)!(2n+1)!)$ :  $a(n)$  is odd exactly when  $n$  is odd and its binary expansion contains no 11. The full valuation is  $v_2(a(n)) = s(n) + s(n+1) - s(3n)$  in binary digit sums. Two independent parity proofs, plus exact valuation distributions, recurrences, asymptotics, extremal formulas and limit laws.

### 50. Polynomial divisibility of height-one factorial ratios

[congruences-and-valuations/factorial-ratio-polynomial-divisibility/article.pdf](#)  
from [bala\\_factorial\\_divisibility\\_research.zip](#)

A general rational-root criterion for uniform polynomial divisibility of balanced integral factorial ratios, applied to Peter Bala's uniform-product conjectures for OEIS A211417 and A295431: explicit multipliers for every product length, with the least multipliers certified in eight particular examples. Includes an atlas of the 52 sporadic height-one ratios with recurrences, generating functions and asymptotic expansions.

### 51. When the Secant Numbers Become Periodic

[congruences-and-valuations/secant-number-periodicity/article.pdf](#)

from secant\_periodicity\_research.zip

Refutes the OEIS A000364 conjecture that the secant numbers from  $n = 1$  are purely periodic modulo every  $m$ : at  $m = 27$ ,  $a(1) \equiv 1$  but  $a(19) \equiv 10$ . The exact replacement is that pure periodicity from index one holds if and only if no odd prime cube divides  $m$ ; the least eventual period always divides  $\varphi(m)$ . Determines the least eventual period and exact onset for every modulus, with densities and a certified mean onset.

## 8 Tetration and digit stabilization

### 52. A Counterexample to a Decimal-Length Bound for Tetration Stabilization

[tetration-and-digit-stabilization/congruence-speed-decimal-length-bound/article.pdf](#)

from tetration\_counterexample.zip

An explicit 2544-digit base  $A = 1 + qu$  ( $q = 5^{2547}$ ) for which  $\nu_{10}(T_{n+1} - T_n) = \min(1 + 2548n, 2547(n + 1))$  at every height, so the congruence speed at height  $\text{len}(A) + 2$  is 2548 while the permanent value is 2547. This contradicts the decimal-length form of Conjecture 1 of Marco Ripà, *The congruence speed formula* (NNTDM 27(4), 2021). The report audits the later Ripà-Onnis confirmation: their weaker valuation bound is compatible with the counterexample, the decimal-length bound is not.

### 53. Tetration, perfect powers, and frozen decimal digits

[tetration-and-digit-stabilization/perfect-power-minima-for-frozen-digits/article.pdf](#)

from tetration\_perfect\_power\_minima.zip

Determines  $\rho(n)$ , the smallest  $n$ th perfect power  $x^n > 1$  whose eventual number of newly frozen terminal decimal digits is exactly  $n$ :  $\rho(1) = 2$ ,  $\rho(2) = 49$ , and  $\rho(n) = B(s)^n$  for  $n \geq 3$  with  $s = n - v_2(n)$  and  $B(s) \in \{2^s \pm 1, 3 \cdot 2^s \pm 1\}$  according to  $s \bmod 4$ . All other roots are ruled out by a deterministic quadratic size barrier plus small exact certificates.

### 54. Phase-shift classification for integer tetration

[tetration-and-digit-stabilization/](#)

[phase-shift-classification/tetration\\_phase\\_classification.pdf](#)

from tetration\_phase\_classification.zip

Proves the 21-word phase-shift conjecture in OEIS A376842, the 14-word even-base restriction from the appendix of Ripà's 2025 paper, and the 28-word prescribed-anchor conjecture in OEIS A376446. The mechanism compares the 2-adic and 5-adic valuations of  $D_b = T_{b+1} - T_b$  at the first permanently constant-speed height: strict 2-adic dominance gives phase 5, strict 5-adic dominance a multiplicative orbit in  $\mathbb{F}_5^\times$ , and balanced valuations exactly four further anchored words.

### 55. A Sharp Convergence Theorem for the $q$ -Newton Series of Subcritical Tetration

[tetration-and-digit-stabilization/q-newton-series-convergence/article/](#)

[tetration\\_qnewton.pdf](#)

from tetration\_qnewton\_research.zip

Proves convergence of the explicit  $q$ -binomial interpolation series of MathOverflow question 259278 for every real base  $1 < a < e^{1/e}$ , with  $\log a = qe^{-q}$ : the grouped series converges absolutely and locally uniformly for  $\text{Re } z > -2$ , converges conditionally on  $\text{Re } z = -2$  except at  $z = -2 + 2\pi ik/(-\log q)$ , and diverges for  $\text{Re } z < -2$ . Adds strict complete monotonicity and uniqueness under an eventual log-convexity condition.

### 56. Stabilization of integer power towers

[tetration-and-digit-stabilization/power-tower-stabilization/article.pdf](#)

from tetration\_stabilization\_research.zip + tetration\_exact\_digits.zip

Refutes the claim that every prime base ending in 9 reaches its permanent decimal congruence speed by height 2 (Ripà, Conjecture 2; arXiv:2208.02622v1, Conjecture 4.1): 2749 is the smallest prime counterexample, with  $H(2749) = 3$ . Proves exact formulas for  $C_a(h)$ ,  $V(a)$  and  $H(a)$ , an exact natural density for onsets, and infinitely many primes at each prescribed onset via CRT

and Dirichlet. Also resolves the three conjectural comments in OEIS A324017 — two proved, the third disproved, with  $3 \uparrow 2 \equiv 27$  and  $3 \uparrow 3 \equiv 59 \pmod{64}$  — via  $T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}$  for even  $q \geq 4$ , so exactly one radix- $q$  digit freezes per height. Adds a factorization-free digit-lifting algorithm, modular saturation across Knuth arrow ranks, and a sign-sensitive local  $p$ -adic Lambert  $W$  formula. *Merged; the shared one-digit-per-height theorem was proved identically in both originals and is proved once here.*

### 57. Delayed digit stabilization: a counterexample and sharp bounds

tetration-and-digit-stabilization/  
iterated-power-digit-stabilization/delayed\_digit\_stabilization.pdf  
from digit\_stabilization\_research.zip

For  $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$  and  $D_a(b) = S_a(b) - S_a(b-1)$ , answers negatively the question whether  $D_a(b) = 1$  for all  $b \geq 3$ : the smallest counterexample is  $a = 3 \cdot 2^{99} + 1$ , with  $S_a(2) = 100$ ,  $S_a(3) = 102$ ,  $D_a(3) = 2$ . Proves minimality, an exact onset formula, the smallest counterexample at every stage, the natural density of exceptions, and arbitrary-radix extensions. The article is explicit that  $a^{10^b}$  is not genuine tetration and derives the tower formulas separately.

## 9 Log-concavity, unimodality and positivity

### 58. Single curves do not force log-concavity

log-concavity-and-unimodality/cluster-variable-log-concavity-counterexample/article.pdf  
from annular\_single\_lamination\_counterexample.zip

An unpunctured annulus with one marked point on each boundary and a single unit-weight simple lamination curve gives, after mutations 2, 1, 2 with the two mutable variables specialised to 1, the polynomial  $1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$ ; the consecutive coefficients 3, 2, 4 form a strict valley, refuting Conjecture 1 of arXiv:2508.04396v3 and conflicting with its general unpunctured-surface unimodality statement. The geometric realisation is proved, not merely the matrix calculation, and a fixed seed gives infinitely many such examples.

### 59. Cycles refute infinite log-concavity for chromatic coefficients

log-concavity-and-unimodality/infinite-log-concavity-of-cycles/article.pdf  
from chromatic\_cycle\_logconcavity.zip

For the cycle  $C_n$  with absolute chromatic coefficients under the iterated operator  $L(a)_k = a_k^2 - a_{k-1}a_{k+1}$ : infinitely log-concave for  $3 \leq n \leq 11$ ; first failure at depth 5 for  $n = 12$ , depth 4 for  $13 \leq n \leq 16$ , and depth 3 for  $n \geq 17$ . So  $C_{12}$  is the smallest cycle counterexample (no minimum over all graphs is claimed), with an explicit third-iterate witness polynomial valid for all  $n \geq 3$ .

### 60. Oscillating Log-Concavity in Geometric Generating Functions

log-concavity-and-unimodality/oscillating-log-concavity-geometric-transform/article.pdf  
from oscillating\_log\_concavity\_research.zip

Challenges 2 and 3 in Section 3 of Heim and Neuhauser, arXiv:2302.13327v1, for the *geometric* transform  $1/(1 - uG(z))$ . For divisor-sum weights, every real exponent and every  $u > 0$  give infinitely many strict log-concavity and log-convexity indices with eventually bounded gaps; likewise for integer power weights with exponent  $\geq 2$ , the exponents 0 and 1 being explicitly exceptional. For square weights at  $u = 1$ , every four consecutive indices carry both signs (sharp), each with density  $1/2$  in every arithmetic progression. A subdominant-pole theorem is the common mechanism.

### 61. A counterexample and a sharp spacing-three criterion for products of $q$ -integers

log-concavity-and-unimodality/q-integer-product-unimodality/report.pdf  
from q\_unimodality\_attack.zip

The necessity half of Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko and Yang (arXiv:2605.12822v1) is false:  $(1+q)^6(1+q^3)$  is symmetric and weakly unimodal although no  $a_i$  is divisible by 3 and  $b = 2$  exceeds the asserted bound, and the example lies inside the source's

claimed range. The report also proves the proposed sufficient condition for every  $r \geq 2$ , a necessary degree bound in the absence of an  $r$ -divisible ordinary factor, and a complete classification of spacings two and three.

## 62. Sharp size thresholds for failures of log-concavity in independence systems

[log-concavity-and-unimodality/independence-system-thresholds/article.pdf](#)

from [sharp\\_log\\_concavity\\_thresholds.zip](#)

For each rank  $r \geq 4$ , determines the smallest ambient ground-set size admitting failure of ordinary, ordered and ultra log-concavity, and shows the three thresholds coincide across all independence systems, those satisfying Xie–Xu’s augmentation bound, and those with additive augmentation parameter 3 (values  $7/6/5$ ,  $7/7/6$ ,  $8/7/7$ , and  $r + 1$  throughout for  $r \geq 7$ ). The initial target, Xie–Xu Conjecture 4.8 (arXiv:2408.09152v1), had already been disproved by Alex Chengyu Li in September 2026; the report states this and refines the target to the rankwise minimum-size question.

## 63. A sharp order-five threshold for higher-order Stirling subset log-concavity

[log-concavity-and-unimodality/higher-order-stirling-rows/article.pdf](#)

from [stirling\\_log\\_concavity.zip](#)

For  $S_{n,k}^{(r)}$ , the number of partitions of  $n + (r - 1)k$  labels into  $k$  blocks of size  $\geq r$ , proves that all rows are log-concave if and only if  $1 \leq r \leq 5$ , with strict inequalities in the interior — the log-concavity assertion of Deb–Sokal Conjecture 1.4(c) (arXiv:2507.18959v1). Also proves the cycle cases  $r = 3, 4$ ; the fifth-order *cycle* conjecture is explicitly left unresolved, and the included operator counterexample is flagged as not being a counterexample to it.

## 64. A sum-of-squares proof of Ulas’s real-zero conjecture

[log-concavity-and-unimodality/stern-polynomial-positivity/stern\\_conjecture\\_note.pdf](#)

from [stern\\_conjecture\\_research.zip](#)

Conjecture 4.4 of Maciej Ulas, arXiv:1909.10844v1, for the Stern polynomials indexed by  $a_n = (4^n - 1)/3$  and  $h_n = 2(4^n - 1)(2 \cdot 4^n + 1)/3 + 1$ : for every  $n \geq 0$  and every real  $t$ ,  $W_n(t) = (A_{n+1}(t) - tA_n(t)/2)^2 + 3t^2 A_n(t)^2/4 > 0$ . The adjacent-polynomial quadratic identity behind it was already in Ulas’s paper and is re-derived, not claimed as new.

## 65. Two-coin counterexamples to generalized Shepp–Olkin thresholds

[log-concavity-and-unimodality/bernoulli-entropy-quasiconcavity/article.pdf](#)

from [bernoulli\\_entropy\\_research.zip](#)

For every finite real  $q > 1$ , both the Rényi- $q$  and the Tsallis- $q$  entropy of a sum of independent Bernoulli variables fail even *quasiconcavity* as a function of the parameter vector, already on the open square  $(0, 1)^2$  with two coins. This disproves the proposed thresholds — Rényi 2 and Tsallis 3.65986... — adjoined to Conjecture 4.2 of arXiv:1503.01570 and restated in arXiv:1510.05390, Section 8.

## 66. Nonreal roots survive in polynomial iterative equations

[log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/article.pdf](#)

from [nonreal\\_roots\\_research.zip](#)

Here  $P[f] = \sum_j a_j f^{\circ j}$  means a combination of *iterates*, not a polynomial in pointwise values. Gives a negative answer to Problem 6.1 of Draga–Morawiec (arXiv:1503.00570), which proposed deleting nonreal maximal-modulus characteristic roots: an explicit odd, strictly increasing, globally bi-Lipschitz homeomorphism realises a cubic whose nonreal root cannot be discarded. Adds a classification of the circular root spectra that occur.

## 67. Canonical binomial decomposition does not preserve real roots

[log-concavity-and-unimodality/binomial-decomposition-real-roots/article.pdf](#)

from [mu\\_welker\\_recursive\\_decomposition.zip](#) + [recursive\\_binomial\\_counterexample.zip](#)

A negative answer to Question 3.10 of Mu and Welker (arXiv:2503.24076v1) for their canonical decomposition  $F(t) = G(t) + tH(t)$  built from the integer binomial expansion of every coefficient: real-rootedness of  $F$  does not pass to the parts. Two cubic witnesses are carried, one with a

repeated root and one with simple roots, each with its own digit table, discriminant pair and exact nonreal zeros; the report also gives the minimal witness within cubics, two all-degree thresholds neither of which implies the other, and a two-sided limit. *Merged; the two originals were two write-ups of one proof program, each already containing both witnesses with the headline and the co-star swapped.*

#### 68. A first-triple criterion for log-concavity of MDS weight distributions

[log-concavity-and-unimodality/mds-weight-log-concavity/article.pdf](#)

from [mds\\_log\\_concavity\\_research.zip](#)

For  $d = n - k + 1 \geq 2$ ,  $k \geq 3$  and  $q > d$ , the compressed formal MDS weight distribution  $(1, A_d, \dots, A_n)$  is log-concave if and only if the single inequality  $A_{d+1}^2 \geq A_d A_{d+2}$  holds — the whole sequence is controlled by its first triple.

#### 69. A uniform Hankel obstruction for higher-order Stirling subset polynomials

[log-concavity-and-unimodality/stirling-hankel-obstruction/article.pdf](#)

from [stirling\\_hankel\\_obstruction.zip](#)

Proves the higher-order negative assertion of Deb–Sokal Conjecture 1.4(d), with a stronger statement at every positive shift: for  $r \geq 3$  and  $a \geq 1$  the  $3 \times 3$  Hankel determinant  $\det(s_{r,a+i+j}(x))$  of the row-generating polynomials fails the required positivity. Clause (d) of that conjecture; clause (c) is the subject of the higher-order Stirling rows entry.

## 10 Graph theory

#### 70. Mutual-visibility sets in Andrásfai graphs

[graph-theory/andrasfai-mutual-visibility/article.pdf](#)

from [andrasfai\\_visibility\\_research.zip](#)

Proves the generating function attributed to Andrew Howroyd in OEIS A391632,  $\sum a(n)x^n = x(4 - 19x + 33x^2 - 40x^3 + 8x^4)/(1 - 10x + 29x^2 - 32x^3 + 8x^4)$ , by an exact weight-preserving bijection between mutual-visibility sets of the  $n$ -Andrásfai graph and anchored closed walks of length  $3n - 1$  in a four-state labelled graph. Adds the full bivariate generating function, an exact binomial formula for every cardinality, maximum-set classification, eventual polynomiality at each fixed deficit, exponential asymptotics, a central limit theorem, and linear-time recognition and exact uniform sampling.

#### 71. A bicyclic 30-vertex graph with domination root $-4$

[graph-theory/domination-root-minus-four-order-30/domination\\_root\\_30.pdf](#)

from [domination\\_root\\_30\\_research.zip](#)

An explicit connected planar graph on 30 vertices and 31 edges, treewidth 2 and domination number 10, with  $D(G, x) = x^{10}(x + 4)Q_{19}(x)$  — a negative answer to question 2 of Section 4 in Alikhani and Griswold, arXiv:2608.00109v1, which asks whether 33 is the minimum order. Their 33-vertex example had already disproved the original integer-root conjecture; no claim of global minimality at 30 is made. Three independent exact polynomial computations agree.

#### 72. A marked-cycle proof of the trapezohedral domination conjecture

[graph-theory/trapezohedral-minimal-dominating-sets/article.pdf](#)

from [trapezohedral\\_domination\\_research.zip](#), [A381190\\_proof\\_package.zip](#)

Proves the generating function labelled conjectural in OEIS A381190: for the  $n$ -trapezohedral graph, the number of dominating sets that are inclusion-minimal among all dominating sets and induce a connected subgraph is  $a_n = 2n(2p_{n-2} + p_{n-3})$  with  $p_m = p_{m-2} + p_{m-3}$  (Padovan). The proof classifies such sets by a cyclic arrangement of gaps of length 2 and 3 with a restricted marked edge. Adds the size distribution, strict log-concavity, an unbiased sampler, a Binet formula and a central limit theorem.

## 11 Enumerative combinatorics

### 73. A Twelve-Value Proof of the A181280 Formula

[enumerative-combinatorics/a181280-binary-matrix-formula/article.pdf](#)  
from [A181280\\_research\\_package.zip](#)

The binary-matrix formula recorded as Conjecture 20 by Kauers and Koutschan (2023) and still marked conjectural in OEIS A181280: an explicit spectral recurrence bound and a finite certificate resting on twelve decisive counts, together with further consequences. `STATUS.md` records that general C-finiteness was already proved in Kauers’s December 2025 lecture slides.

### 74. Esthetic Numbers, Folded Paths, and Algebraic Diagonals

[enumerative-combinatorics/esthetic-numbers-and-folded-paths/article.pdf](#)  
from [esthetic\\_walks\\_research.zip](#)

The diagonal conjecture  $T(q, q-2) = A182555(q-2)$  in OEIS A377000, where  $T(q, k)$  counts base- $q$  positive integers of length  $k$  whose adjacent digits differ by one. Proves it exactly, proves the entry’s other two conjectures as retrieved, and derives formulas for every fixed-offset diagonal  $q = k + d$ .

### 75. Diagonal Hardinian Arrays: A Fixed-Parameter Asymptotic Formula

[enumerative-combinatorics/hardinian-array-diagonal-asymptotics/article.pdf](#)  
from [hardinian\\_arrays\\_research.zip](#)

For every fixed  $r \geq 0$ ,  $H_r(n, n) \sim C_r 4^{rn} n^{-r(r-1)/2}$  with an explicit  $C_r$  in Gamma values. The formula confirms the constants tabulated by Dougherty-Bliss and Kauers (2024) but refutes their proposed restriction to powers of 2, 3 and  $\pi$ :  $C_7 = 2^{18} 5^2 / (3^{45} \pi^3)$ . Adds an exact finite determinant at fixed array size, an exact enumeration algorithm, and a limiting negative-binomial law for total deficits.

### 76. An explicit formula for lucky parking spots

[enumerative-combinatorics/lucky-parking-spots/article.pdf](#)  
from [lucky\\_parking\\_spots\\_research.zip](#)

Proves Conjecture 13 of Butler, Hadaway, Lenius, Martens and Moats (JIS 29 (2026), Article 26.1.1): a closed form for  $C(n, j)$ , the number of parking functions in which *spot*  $j$  is lucky. Adds an exact reflection law, all fixed right-edge columns, a closed form for OEIS A374533, Poisson edge limits, uniform bulk convergence, polynomial reciprocity and asymptotic expansions.

### 77. A hexagonal-lattice proof of the A195806 conjecture

[enumerative-combinatorics/a195806-hexagonal-lattice/article.pdf](#)  
from [A195806\\_hexagonal\\_proof.zip](#)

An elementary derivation of the six residue-class formulas and the constant-coefficient recurrence recorded as conjectural in OEIS A195806, starting from the original triangular-array constraints rather than from interpolation. The steps are an integral six-parameter bijection, four independent translation intervals, a dihedral chamber reduction, and an exact lattice count.

### 78. Leaves of large type in the numerical-semigroup tree

[enumerative-combinatorics/numerical-semigroup-leaf-types/article.pdf](#)  
from [leaf\\_type\\_counterexample.zip](#)

Disproves Conjecture 4.4 of Chappelon, Ramírez Alfonsín and Stamate, *Numerical semigroup tree: type-representation* (arXiv:2507.15006v2), with an explicit genus-34 counterexample, and establishes a sharp asymptotic order for the type of leaves in that tree.

### 79. Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

[enumerative-combinatorics/adjacency-bounded-132-avoiders/article.pdf](#)  
from [strict\\_growth\\_132.zip](#)

For 132-avoiding permutations with every adjacent difference at most  $m$  and growth constant  $\alpha_m$ : proves  $\alpha_{m+1} > \alpha_m$  for all  $m \geq 1$  — the remaining strict-inequality part of Nadler’s Conjecture 2 —

and the refinement  $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$ , with  $2 \log \pi - 4 \log 2 \leq \liminf E_m \leq \limsup E_m \leq 2 \log \pi$ . The all- $m$  component ordering and convergence of  $E_m$  are explicitly not claimed.

## 80. Throwback Dynamics with Repeated Entries

[enumerative-combinatorics/throwback-queue-dynamics/throwback\\_research.pdf](#)  
 from [throwback\\_research.zip](#)

For the queue process that removes the head and reinserts it at the zero-based position equal to its weight: every token recurs if and only if at most  $m$  input tokens have weight at most  $m$ , for every  $m \geq 1$ , with an explicit description of the recurrent tokens when that fails. Adds exact cycle periods and frequencies, a coprimality criterion for perfect mixing, Catalan and prime-parking counts for recurrent cores, and the largest labelled period of a core of size  $r$ .

## 81. Reciprocity and Matrix Duality for Preorder Polytopes

[enumerative-combinatorics/preorder-polytope-reciprocity/article.pdf](#)  
 from [preorder\\_reciprocity.zip](#)

Proves Conjectures 7.2, 4.2 and 4.4 of Athanasiadis and Chapoton, *Polytopes and posets associated to preorders* (arXiv:2605.26916v1), and the ordinary  $q = 1$  case of their Conjecture 4.9 (not its full  $q$ -refinement). Two mechanisms: Ehrhart–Macdonald reciprocity after subtracting the all-ones vector, extending the double identity to independently weighted capacities; and duplication of both colour classes of the bipartite graph followed by Postnikov’s lattice-point formula, giving the conjectured matrix transpose.

## 82. Support reciprocity and the $q$ -zeta polynomial of preorder lattice-point posets

[enumerative-combinatorics/preorder-q-zeta-reciprocity/article.pdf](#)  
 from [preorder\\_qzeta\\_reciprocity.zip](#) + [preorder\\_q\\_zeta\\_reciprocity.zip](#) +  
[preorder\\_q\\_reciprocity\\_research.zip](#)

Proves the full  $q$ -refinement of Athanasiadis–Chapoton Conjecture 4.9, equation (21): for a preorder  $\tau$  on  $[n]$ , with  $P_\tau$  the nonnegative integer vectors satisfying  $a(I) \leq |I|$  on every order ideal and  $B_\tau$  its maximal elements,  $Z_q(P_\tau, [-1]_q) = (-1)^n \sum_{b \in B_\tau} q^{-|\text{supp}(b)|}$ , together with the stronger identity at each exact support. The  $q = 1$  case was already covered by the preorder-polytope reciprocity report and is cited, not reclaimed. *Merged from three reports that reached the same two theorems by three architecturally independent routes — an incidence inversion with Boolean localization, the same with a Hall-based privatization and a self-contained Euler identity, and a capacity-enumerator reciprocity with a transfer involution — and all three survive.*

## 83. An explicit shelling for preorder lattice-point posets

[enumerative-combinatorics/preorder-shellability/article.pdf](#)  
 from [preorder\\_shellability\\_research.zip](#)

An affirmative solution to Question 4.6 of Athanasiadis and Chapoton (arXiv:2605.26916v1, p. 13): an explicit pure shelling of the order complex of  $P_\tau$  with its minimum removed, and a coning that shells the order complex of  $P_\tau$  itself. Both complexes are therefore Cohen–Macaulay over the integers and over every field. A different question of that paper from the three other preorder entries.

## 84. Multiple-chain preorder polytopes

[enumerative-combinatorics/multiple-chain-exponential-formula/article.pdf](#)  
 from [multiple\\_chain\\_exponential\\_formula\\_research.zip](#) + [multiple\\_chains\\_research.zip](#)

Proves the exponential generating-function identity proposed in Section 9.5 of Athanasiadis and Chapoton (arXiv:2605.26916v1) for lattice points of multiple-chain preorder polytopes counted by number of nonzero coordinates, in a rectangular generalization with independent block width and height. Adds a duality corollary, an Ehrhart evaluation, a degree- $2^d$  root theorem with a quartic elimination certificate, two treatments of the  $\gamma$ -vector, a Stieltjes moment representation, the boundary constant in two forms, and a central limit theorem with exact variance. *Merged; the identity is proved twice, once through a subset encoding with a bounded symbol and once through slack meanders in a Laurent ring, and the eleven deliberate redundancies are listed in the text.*

### 85. Rationality and growth of stabilized permutation-weight series

[enumerative-combinatorics/stabilized-permutation-weights/article.pdf](#)

from [stabilized\\_permutation\\_weights.zip](#)

Studies the full stabilized generating functions of the permutation-weight  $q$ -Eulerian polynomials, beyond their initial connection with OEIS A256193: an all- $d$  finite-spectrum theorem, rationality, a universal integer denominator, the exact radius of convergence and residue-class asymptotics. The two-descent sequence gets an explicit formula in powers of 3, powers of 2 and Fibonacci numbers.

### 86. A Reflexive Root-Polytope Model for Preorder $h$ -Polynomials

[enumerative-combinatorics/preorder-root-polytopes/article.pdf](#)

from [preorder\\_root\\_polytopes.zip](#)

Proves Conjecture 5.1(c) of Athanasiadis and Chapoton (arXiv:2605.26916v1): for a preorder  $\tau$  on  $[n]$  the directed root polytope  $C_\tau = \text{conv}(\{\pm e_i\} \cup \{e_i - e_j : j \leq_\tau i\})$  is reflexive with  $h^*(C_\tau, t) = h_\tau(t)$ , and any pulling triangulation of its boundary is combinatorially the boundary of a simplicial polytope with that  $h$ -polynomial. Palindromicity and unimodality, Conjecture 5.1(a,b), follow, as do the full  $g$ -theorem conditions. The higher-dimensional augmented bipartite identity and preorder duality are credited to Dai, Hou, Liu, Thawinrak and Wang (arXiv:2608.16037v2); flag realizability,  $\gamma$ -positivity and real-rootedness are not proved. A different conjecture of the same paper from the preorder-polytope reciprocity report.

### 87. Largest Simplex Faces under Ordinal Sums

[enumerative-combinatorics/](#)

[simplex-faces-of-order-and-chain-polytopes/Largest\\_Simplex\\_Faces.pdf](#)

from [Series\\_Parallel\\_Simplex\\_Faces\\_Research.zip](#)

Answers the concluding open question of Freij-Hollanti and Lundström, *Simplex inequalities of order and chain polytopes of recursively defined posets* (Order 43 (2026), article 31). For  $s_O(P)$  and  $s_C(P)$ , the largest dimensions of simplex *faces* of the order and chain polytopes: both are computable in  $O(n)$  on any series-parallel decomposition, the order case carrying eight normalized endpoint states. For an ordinal sum of antichains,  $s_O = r + \nu$  with  $\nu$  a maximum  $b$ -matching of the layer path under capacities  $\min(a_i - 1, 2)$ , and  $s_C = r + \max(0, k - 1)$ . The extremal gap is sharp:  $0 \leq s_C - s_O \leq \lfloor (n - 2)/3 \rfloor$ , attained for every  $n \geq 2$ .

### 88. Skew partitions, continued fractions and connected components

[enumerative-combinatorics/skew-partition-continued-fractions/article.pdf](#)

from [A225114\\_proof\\_bundle.zip](#)

Proves the repeated-power continued-fraction formula recorded as conjectural in OEIS A225114, which counts skew partition diagrams with no empty row or column. Unique concatenation gives  $A(q) = 1/(1 - P(q))$  for the edge-connected diagrams, and a boundary-path encoding followed by a weighted two-coloured Motzkin-path decomposition gives  $P(q) = q/(1 - 2q - q^3/(1 - 2q^2 - q^5/(1 - 2q^3 - \dots)))$ , whence  $A(q) = 1/(2 - C(q))$  with  $C = 1 + P$ . Adds a four-variable refinement, a formula by height, width and component count, a sharp truncation bound, exponential asymptotics and a central limit theorem for the number of components. **Explicitly not a novelty claim:** a prior-art check found Pasechnik's MathOverflow sketch of ten September 2024, and the entry credits Kurkov; the report says so first and credits both.

### 89. From mesh avoidance to Catalan inflation

[enumerative-combinatorics/mesh-avoidance-catalan-inflation/article.pdf](#)

from [A289587\\_proof.zip](#), [OEIS\\_A289587\\_proof.zip](#)

Proves Scheuerle's generating-function conjecture in OEIS A289587 for permutations avoiding 321 and the mesh pattern  $(12, 174)$ , and the equal count for  $(12, 234)$  by inversion: with  $D = x^4 - 2x^3 - 5x^2 - 2x + 1$ ,  $P = x^4 + 4x^3 + 11x^2 + 10x + 3$  and  $Q = x^2 + 5x + 3$ , the generating function is  $(P - Q\sqrt{D})/(8x(1+x)^2)$ . The headline structural result is the exact *occurrence count*  $\text{occ} = \binom{f}{2} + \sum_U \binom{|U|}{2}$ , of which avoidance is the case  $\text{occ} = 0$ . Merged from two reports that reached the identical generating function by the identical run-contraction substitution  $u \mapsto u/(1+xu)$ ;

the merge keeps *both* refinements, because they refine the same sequence by different statistics — excedances and records — and are genuinely different theorems, agreeing only in total. The two local predicates they are built on disagree on 31 of the 321-avoiders of length at most six, the smallest being 12, and the merged article proves where they coincide rather than assuming it.

## 90. Prescribed coefficient-lattice minima

[enumerative-combinatorics/coefficient-lattice-minima/article.pdf](#)

from [coefficient\\_lattice\\_minima\\_research.zip](#)

Settles the unnumbered realization conjecture after Lemma 1 of O’Bryant, *On Nathanson’s Triangular Number Phenomenon* (arXiv:2506.20836): every nondecreasing list of even integers is the list of successive  $L^1$ -minima of the coefficient lattice of some finite integer set. The witness is the mixed-radix set  $A = \{0, 1, h_1, h_1 h_2, \dots\}$ , whose minima are exactly  $(2h_1, \dots, 2h_r)$  with  $r + 2$  distinct elements, and the radices may be given in any order.

## 91. Coefficientwise magic positivity through dimension ten

[enumerative-combinatorics/magic-positivity-parking-polytopes/article.pdf](#)

from [magic\\_positivity\\_research.zip](#)

A polytope is *magic positive* when its Ehrhart polynomial has nonnegative coefficients in the basis  $t^j(t+1)^{n-j}$ . Proves, in exact arithmetic, that the generalized parking-function polytope  $X_n(b)$  is magic positive for every dimension  $3 \leq n \leq 10$  and *every* positive integer parameter vector, uniformly in  $b$  rather than on a sampled box — a partial solution of Conjecture 8.1 of Hill, Luo, Trinh and Vindas-Meléndez (arXiv:2607.15503v1). Dimension three is proved by printed algebraic formulas; the first-coefficient harmonic-number formula and the endpoint identities hold in every dimension. The conjecture in unrestricted dimension remains open.

## 92. Monotone blocks and reverse replies

[enumerative-combinatorics/reverse-reply-avoidance-game/article.pdf](#)

from [reverse\\_reply\\_solution.zip](#)

Proves Conjecture 2.9 of Ulfarsson (arXiv:2603.16004): for every  $k \geq 3$  and every  $n \geq R(k) = k^4 - 2k^3 - 3k^2 + 5k + 5$ , the second player wins the normal-play permutation-avoidance game on  $S_n$  with pattern length  $k$  by always replying with the reversal, so the Sprague–Grundy value is 0. The proof is elementary, with no computer search: a minimal-support lemma isolates one pattern inside a family built from singletons plus one inflatable monotone interval, and inflation restores the original length so the reply is legal.  $R(k)$  is asserted sufficient, not minimal.

## 93. Nonintegral tropical degree from two horizontal segments

[enumerative-combinatorics/tropical-hilbert-nonintegral-degree/article.pdf](#)

from [tropical\\_hilbert\\_counterexample.zip](#)

Computes the strict-witness tropical Hilbert function of Grigoriev (arXiv:2404.06440v2) exactly for a two-segment set:  $\text{TH}_{V_d}(k) = k + \min\{k, \lceil k/d \rceil\} + 1$ . Taking  $d = 3$  gives tropical degree  $4/3$  — not an integer — and a quasipolynomial of least period 3, so eventual polynomiality fails. Every rational in  $(1, 2]$  is realized, denominators are unbounded, and for  $d = p/q > 1$  in lowest terms the least eventual period is exactly  $p$ . Both bounds are proved by hand.

# 12 Generating functions and asymptotics

## 12.1 Coefficient counts of binary products

Three reports on counting coefficients of  $\prod_i (1 + x^{G_i})$ . The first two answer, independently, the universal-rationality part of Richard Stanley’s MathOverflow question 431075 (“A conjectured rational generating function”, 2022), which had no posted answer when checked on 19 September 2026. A third answer used the same mixed-base construction and the same enumerative argument as the first, and has been merged into it.

## 94. A Mixed-Base Counterexample to Stanley’s Rationality Question

generating-functions-and-asymptotics/binary-product-coefficient-counts/  
mixed-base-counterexample/article.pdf  
from stanley\_mixed\_base\_counterexample.zip + Stanley\_Rationality\_Counterexample.zip

With  $G_{2j+1} = 2^j$  and  $G_{2j+2} = 3^j$  the exponent generating function is rational, while  $\nu(2n) = 2^{n - \lfloor (n+1)\alpha \rfloor}$  for  $\alpha = \log_3 2$  counts the coefficients equal to one in  $\prod_{i \leq N} (1 + x^{G_i})$  and its generating function is not rational, not D-finite, not P-recursive, and has a natural boundary. Classifies rationality in the family interleaving  $a^j$  and  $q^j$  with digit polynomial  $1 + y + \dots + y^{a-1}$  for all  $2 \leq a < q$  — Stanley's general  $r = a - 1$  — as holding exactly when  $q^u = a^v$ . Adds a finite-quotient rigidity theorem, an abstract natural-boundary theorem for  $\sum q^{\lfloor n\theta \rfloor} t^n$  at every irrational  $\theta$ , the exact limit set of  $a_n/\lambda^n$ , and a streaming recurrence making the point that non-P-recursive does not mean hard to compute. *Merged; the enumerative half was the same argument in both originals, but the two routes to non-rationality differ and both are kept — an elementary finite-state reduction mod  $p$ , and the rigidity route that reaches the stronger transcendence conclusions.*

## 95. Rational exponents, nonrational coefficient counts

generating-functions-and-asymptotics/binary-product-coefficient-counts/  
recursive-exponent-construction/research.pdf  
from stanley\_counterexample.zip

A different negative answer, from the recursion  $a_0 = 2$ ,  $a_1 = 1$ ,  $a_n = a_{n-1} - 2a_{n-2}$ , with  $b_n = T_{n-1} + 2^n - 1 + a_n$ ,  $T_n = T_{n-1} + b_n$ ,  $G_{2n-1} = 2^{n-1}$  and  $G_{2n} = b_n$ : the counting series has a natural boundary on the unit circle, hence is neither rational, algebraic nor D-finite.

## 96. Coefficient residues in generalized Fibonacci products

generating-functions-and-asymptotics/binary-product-coefficient-counts/  
kbonacci-coefficient-residues/article.pdf  
from stanley\_kbonacci\_research.zip + quasifibonacci\_parity\_research.zip

Settles Conjectures 6.2 and 6.3 of Stanley, *Theorems and Conjectures on Some Rational Generating Functions* (arXiv:2101.02131v3, pp. 23–24). For  $k$ -bonacci seeds each exceeding the sum of their predecessors, the number of odd coefficients of  $\prod_{i \leq n} (1 + x^{w_i})$  has generating function  $(1 + 2z^k)/(1 - 2z + 2z^k - 2z^{k+1})$ , independently of the seeds; the proof builds periodic signs of block length  $k + 1$  and block product  $-1$  and shows every signed finite product is flat. *Merged; the Conjecture 6.3 halves proved the same statement on the same domain, and the surviving report adds Conjecture 6.2 and the asymptotic analysis.*

## 12.2 Single-sequence OEIS studies

Nine reports on the growth or the closed form of a single OEIS sequence. Three of them count the terms in the derivatives of a power tower — of  $x^{x^x}$ ,  $x^{x^2}$  and  $x^x$  — and reach three different answers by three different recurrences; two more share a Bell-scale saddle-point method on unrelated sequences. Shared technique is not shared theorem, and none of these nine duplicates another.

## 97. Bell-scale growth of OEIS A088714

generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
a088714-bell-scale-growth/article.pdf  
from A088714\_asymptotic\_growth.zip

For the unique formal series with  $A(x) = 1 + xA(x)^2A(xA(x))$ , proves  $\log a_n = n \log n - n \log \log n - n + O(n \log \log n / \log n)$ , so  $(e \log n / n) a_n^{1/n} \rightarrow 1$  and the ordinary generating function has radius of convergence zero while the exponential one is entire. The lower bound comes from a single explicit path through an exact triangular recurrence, the upper from a Stirling-positivity comparison with derivatives of Touchard polynomials. The multiplicative equivalent is *not* claimed, and the finer Bell-number form is labelled a conjecture with a sufficient ratio condition isolated.

## 98. The exponential growth constant of OEIS A158415

generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
a158415-growth-constant/article.pdf

from A158415\_asymptotic\_growth.zip

For the number of distinct values expressible with  $n$  symbols from  $E ::= 1 \mid \sqrt{E} \mid (E + E)$ , proves a computable  $\gamma > 1$  exists with  $a(n)^{1/n} \rightarrow \gamma$  and  $1.86204 < \gamma < 1.9208$ , gives an effective characterisation as a supremum of algebraic finite-data bounds, and proves the boundary identity  $\sum a(n)\gamma^{-n} = \gamma^2$ , which rules out a pure  $C\gamma^n$  prefactor. The rigorous lower bounds rest on a uniform injectivity theorem: any finite alphabet of representable values can serve as nested-radical digits, via an odd valuation at a prime above 2. No closed form for  $\gamma$  is claimed, and terms 22–28 get certified intervals rather than exact values.

### 99. An exact coefficient formula for Ramanujan’s theta function

generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
a260306-formula-proof/a260306\_proof.pdf  
from A260306\_formula\_proof.zip

Proves both finite formulas recorded in OEIS A260306, for every coefficient index, for the expansion  $\theta(N) \sim \sum c_n N^{-n}$  defined by  $\sum_{k < N} N^k/k! + \theta(N)N^N/N! = e^N/2$ . The route is an exact Laplace-integral difference, an analytic change of variable with  $F(\psi(w)) = w^2/2$ , and Lagrange inversion reducing each  $c_n$  to one coefficient extraction, then a finite change of basis into the entry’s double and triple sums. Includes the analytic error estimate the identification needs, rather than only checking the two sums agree, and identifies A065973 as the reduced denominators.

### 100. The maximal positive coefficient of a Hermite polynomial

generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
a277280-hermite-maximum/article.pdf  
from A277280\_asymptotics.zip

An exact closed form for OEIS A277280, the largest coefficient of the physicists’ Hermite polynomial  $H_n$  with signs retained: with  $k_n = \lfloor (2n + 6 - \lfloor \sqrt{8n + 21} \rfloor)/8 \rfloor$  and  $d_n = n - 4k_n$ , the exponent  $d_n$  is the unique maximizer and  $a(n) = n! 2^{d_n} / ((2k_n)! d_n!)$ . The distinction from the largest coefficient in absolute value is kept explicit throughout.

### 101. Asymptotics of OEIS A277364

generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
a277364-bell-asymptotics/A277364\_asymptotics.pdf  
from A277364\_asymptotics.zip

For  $a(n) = \sum_{k \leq n/2} S(n, k)$ , partitions of an  $n$ -set into at most  $\lfloor n/2 \rfloor$  blocks, proves  $a(n) \sim B_n$  with an explicit first correction: writing  $r = W_0(n)$ ,  $a(n) = \exp\{n(r - 1 + 1/r) - 1\}(1 + r)^{-1/2}(1 - P(r)/n + O(r^2/n^2))$  for an explicit rational  $P$ . Also gives a parity-split equivalent for the omitted tail  $B_n - a(n)$ .

### 102. Term counts in the derivatives of power towers

generating-functions-and-asymptotics/oeis-sequence-asymptotics/  
power-tower-derivative-term-counts/article.pdf  
from A293239\_research\_report.zip, A290268\_partial\_results.zip,  
A281434\_exact\_algorithm\_and\_cubic\_growth.zip

One article on three OEIS sequences that ask the same question of three functions: how many distinct monomials are in the fully collected  $n$ th derivative of  $x^x$  (A293239),  $x^{x^2}$  (A290268) and  $x^{x^x}$  (A281434)? Part I proves once the apparatus all three had established separately — the canonical differential-polynomial form  $f^{(n)} = fP_n$  or  $fx^{-n}P_n$  with its first-order integer recurrence, the independence of logarithmic monomials that makes the count an invariant of  $f$  rather than of a differentiation algorithm, the identity  $a(n) = |\text{supp } P_n|$ , and the envelope bound. Two normalizations are needed, because neither clause covers all three in the frames the parts use. **None of the three headline questions is fully settled, and the article says so in its abstract:** for  $x^x$  the count reduces exactly to zeros of the first-kind Lehmer–Comtet triangle A008296 and  $a(n) = \Theta(n^2)$ , but the closed formula needs an isolated nonvanishing step — and separately the recurrence printed in the entry for  $n > 6$  is *disproved* at  $n = 8$ ; for  $x^{x^2}$  the conjectured expression is proved to be an upper bound and  $a(n) = \Theta(n^2)$ , but not the constant  $\frac{1}{2}$ ;

for  $x^{x^x}$  explicit bounds give  $a(n) = \Theta(n^3)$  but not  $a(n) \sim \frac{2}{3}n^3$ , and there the obstruction is not a nonvanishing statement but a density one, since further zeros provably exist. A closing part sets the three side by side and records what they share beyond the framework: every cancellation any of them can prove is a parity zero of the same falling factorial. This is a thematic consolidation, not a deduplication: the three were checked and are not duplicates, and all fifty numbered results of the three originals are preserved unchanged.

### 103. Iterated sums and products from one

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a352969-correctness-and-growth/article.pdf](#)  
 from A352969\_correctness\_and\_growth.zip

For  $S_0 = \{1\}$ ,  $S_{n+1} = (S_n + S_n) \cup (S_n \cdot S_n)$  and  $a_n = |S_n|$  (OEIS A352969), proves correctness of the listed terms and the doubly exponential law  $2^{2^{n-rn-4}} \leq a_n \leq 2^{2^{n-1}}$ , hence  $\log \log a_n = n + O(\sqrt{\log n})$ . The step is simultaneous, not repeated closure — a distinction the report is careful about.

### 104. Distinct third derivatives of a power tower

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a199085-exponentiation-derivative-values/article.pdf](#)  
 from A199085\_generating\_function.zip

For OEIS A199085, the number of distinct values taken by the third derivative at  $x = 1$  over all fully parenthesized exponentiations of  $n$  copies of  $x$ , proves  $a(1) = a(2) = 1$  and  $a(n) = \lfloor (n^2 - 2)/3 \rfloor$  for  $n \geq 3$ , with the rational generating function  $z/(1-z) + z^3/(1-z)^3 - z^7/((1-z)^2(1-z^3))$ . The proof is for all  $n$ , not an extrapolation: every attainable pair of second and third derivatives is classified and a witness constructed for each admissible pair. Despite the shared vocabulary this is a different question from *power-tower-derivative-term-counts*, which fixes the function and varies the derivative order; here the order is fixed at three and the parenthesization varies.

## 12.3 Asymptotics, records and discrepancy

### 105. The Exponential Scale of Gregory Sign Thresholds

[generating-functions-and-asymptotics/gregory-coefficient-sign-thresholds/gregory\\_sign\\_threshold.pdf](#)  
 from gregory\_sign\_threshold\_research.zip

For  $(x/\log(1+x))^m = \sum_n G_n^{(m)} x^n$  and  $\text{Gr}(m)$  the least  $N$  beyond which  $(-1)^{n-1} G_n^{(m)} > 0$ , proves  $\log \text{Gr}(m) = m + 1 - \gamma - (\zeta(2) + \zeta(3))/m + C_2/m^2 + O(m^{-3})$  with  $C_2$  explicit, plus a finite algorithm for every coefficient of the complete expansion; in particular  $\text{Gr}(m)/e^m \rightarrow e^{1-\gamma}$  and  $\text{Gr}(m)/3^m \rightarrow 0$ . This replaces Conjecture 6.1 of Xu and Zhao (arXiv:2609.11072v1), whose printed lower bound already conflicts with its own reported  $\text{Gr}(5) = 113$ ; that value is independently certified here.

### 106. Exact Records in the Square-Root-of-Three Walk

[generating-functions-and-asymptotics/sqrt-three-walk-records/article.pdf](#)  
 from sqrt3\_record\_recurrence.zip

Proves the four-phase record recurrence for  $S(N) = \sum_{n \leq N} (-1)^{\lfloor n\sqrt{3} \rfloor}$  in the corrected form on page 7 of Bruin and Fokkink, *On the records and zeros of a deterministic random walk* (arXiv:2503.11734v2), where it is experimental and attributed to Arias de Reyna and van de Lune. Includes minimality of every claimed first passage, a rational generating function, closed forms, sharp logarithmic extrema, and an extension to every even integer trace  $D \geq 4$ .

### 107. Sharp discrepancy for a family of binary substitutions

[generating-functions-and-asymptotics/binary-substitution-discrepancy/article.pdf](#)  
 from oeis\_discrepancy\_counterexamples.zip

For the fixed word of  $0 \mapsto 1$ ,  $1 \mapsto 101010$ , the position sequences A284365 and A284366 both carry a conjectured upper position error below 2, and both conjectures are false: the first counterexamples

are at ranks 2,977,771 and 8,933,313, with common error  $(2977771\sqrt{21} - 13645857)/2 \approx 2.00487$  and sharp common upper bound  $(6 + \sqrt{21})/5 \approx 2.11652$ . Proves complete discrepancy limit intervals for every substitution  $0 \mapsto 1$ ,  $1 \mapsto (10)^m$ , with infinite counterexample families and exact recurrences for their indices.

### 108. Exact eventual maximizers of Rudin–Shapiro autocorrelations

[generating-functions-and-asymptotics/](#)

[rudin-shapiro-autocorrelation-maximizers/article.pdf](#)

from [rudin\\_shapiro\\_maximizers.zip](#)

For the aperiodic autocorrelation  $C_m(k)$  of the level- $m$  Rudin–Shapiro sequence, classifies the maximizing shift exactly for all large  $m$ :  $k_m^* = (2^{m+1} + (-1)^m)/3$ , with  $A_m = b_{m-2}$ , settling the discrete uniqueness-and-location conjecture of Tarnu (J. Approx. Theory 287 (2023) 105866), repeated as Conjecture 1.5 of Choi–Tarnu (2025), including the limit  $k_m^*/2^m \rightarrow 2/3$ . The certificate is a rational polytope argument with exact arithmetic.

### 109. Apéry arrays and conjectural zeta accelerations

[generating-functions-and-asymptotics/apery-array-zeta-accelerations/article.pdf](#)

from [Apery\\_OEIS\\_Conjectures\\_Proofs.zip](#), [apery\\_oeis\\_conjectures.zip](#)

Proves the entire shifted-diagonal  $\zeta(3)$  family recorded as conjectural in OEIS A143007, and both the super- and subdiagonal  $\zeta(2)$  families of A108625, at every nonnegative offset: for all  $k \geq 0$ ,  $\zeta(3) = H_k^{(3)} + \sum_{n \geq 1} (2n+k)(3n^2+3nk+k^2)/(n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k})$ . The main diagonals are the classical Apéry sequences A005259 and A005258. Proofs are finite telescoping identities with rational lattice potentials and explicit limit estimates; no computer test or unproved conjecture enters. Adds rational error bounds, numerator and denominator recurrences, a three-term recurrence along every shifted diagonal, and sharp fixed-offset asymptotic constants. Merged from two reports whose boxed identities were character-identical up to macro names.